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NSF SBIR PHASE I PROPOSAL

Solicitation NSF 26-510 · submitted via Research.gov · gated by an approved Project Pitch (eight-digit case number entered here)

Field	Entry
Small business concern	∞
Authorized Company Officer	Catherine Robinson-Rutella, Founder and Chief Architect
Project title	A Human-Experience Humanoid Platform for the Home: Feasibility of Home-Capable Humanoids at One-Third the Cost of Commercial-Grade Systems
SBIR topic code	Robotics (confirm exact code against the live dropdown at submission)
Project Pitch case number	[eight-digit number - issued after the pitch is approved]
Award requested	$224,700 (mock arithmetic below; ceiling $305,000)
Project duration	9 months (allowed range 6-18)
Company eligibility	US small business, under 500 employees, majority US-citizen owned, no venture majority ownership - attested "small business" by the founder, Oct 23, 2026
Address / DUNS-UEI / registrations	[New York address · SAM.gov UEI issued by registration · Research.gov ID]

PROJECT SUMMARY

Overview

∞ will prove the technical feasibility of a humanoid robot designed for the one environment no humanoid manufacturer serves: the home. Over 140 companies build humanoids and roughly 7,000-8,000 commercial units operate today, nearly all aimed at factories; no humanoid on the market in 2026 is ready for home use. The project validates the four highest-risk subsystems of an existing, complete, costed full-body specification - actuation within home-safety limits, truthful-thermal skin, power and thermal management, and self-repair materials - at a target bill of materials of one-tenth to one-third the cost of a commercial-grade humanoid ($200K class).

Technical merit

The innovation is the organ-ledger design method: every subsystem specified, costed, and traded off in a version-controlled ledger so the platform is designed as a coherent body rather than assembled as a parts list. Feasibility is unproven at project start; Phase I converts a 17-section engineering specification (26 numbered design laws, costed BOM $19,193-63,639) into a validated, test-evidenced feasibility case.

Commercial merit

The entry wedge is care: people with disabilities and aging-in-place households, a market the field does not touch. A humanoid priced against annual in-home care costs rather than industrial budgets opens a new market category. Phase I's deliverables de-risk the Phase II prototype that anchors that claim.

PAGES 3-9 · PROJECT DESCRIPTION (10 PAGES MINIMUM, 15 MAXIMUM UNDER NSF 26-510 - THE MOCK SHOWS THE SECTION SHAPE; THE REAL DRAFT EXPANDS IT)

PROJECT DESCRIPTION

Identity and Innovation

Who we are, what the innovation is, and why NSF should fund it: the organ-ledger design method, human-experience requirements as engineering requirements, and radical cost discipline - the three claims carried by the field census, the costed BOM, and the version-controlled record (707 pressed documents, 25 original research papers, over 1,240 commits, every hash verifiable).

Technical Objectives, Approach and Work Plan

This header is REQUIRED by the solicitation instructions. Four feasibility objectives, each with a milestone and success criterion:

Task	Objective	Approach	Milestone (month)	Success criterion
T1	Actuation & mobility budget	Bench-test one joint module against the specification's force and safety limits	M3	Working module at or under its ledgered cost target; full-body actuation budget with arithmetic shown
T2	Thermal & skin layer	Fabricate and cycle a skin coupon at human-range thermal output	M5	Coupon survives the spec's durability cycle count, costed per unit area
T3	Power & thermal management	Simulate the endocrine-inspired regulatory architecture over a full-day home care duty cycle	M7	Duty cycle stays inside all home-safety margins, measured data substituted where available
T4	Self-repair & maintainability	Test sealant/coating/clotting-analog compatibility matrix; one bench demo	M9	Compatibility report and demonstration, folded into the revised BOM

The real draft also carries: the key challenges and why they are open (cost-performance cliff, e-skin durability, power density, body-level integration), a related-work map against the 140-company field, a personnel-level schedule, and the PI's 20 hours per week named in the plan.

Commercial Impact

The home and care market, the wedge population, the right to win (costed design, the method, the execution record), the business model (platform licensing, direct sale priced against human care costs), and the Phase II path to a prototype. This section absorbs the market data already gathered for ZYK-410/411.

Broader Impacts

This header is also REQUIRED. The draft: disability and aging-in-place impact, a design methodology publishable for the field, workforce development through the specialist hires, and the housing-and-care thesis the founder's practice is built on.

REFERENCES CITED

Carried whole from the company's existing artifacts: the 291-source bibliography compiled by script from the build corpus (specification v3.0), the 14-reference academic register of the collaborator paper (ZYK-411), and the October 2026 web-verified field census. In the real draft these merge and dedupe into one numbered list - roughly 40-60 entries, every one with a verifiable source.

BIOGRAPHICAL SKETCH - CATHERINE ROBINSON-RUTELLA (PI)

Professional preparation. Physics; Fordham University, psychology coursework (2024-2025).

Positions. Founder and Chief Architect, ∞ (2026-present); independent research program, Stardust Lab (2026-present); ten years full-cycle business development before the lab.

Products most relevant. 25 original research papers, the 17-section humanoid engineering specification, the costed bill of materials and feasibility arithmetic, the collaborator paper - all version-controlled and verifiable commit by commit.

Synergistic activities. The organ-ledger design method; the design corpus published under the pen name Cici Pink.

> MOCK NOTE: the real biosketch follows NSF's current format rules exactly (sections, page count) - checked against the PAPPG at drafting time. Published-name handling (Cici Pink) is the founder's call.

BUDGET (9-MONTH PHASE I)

Line	Item	Amount
A/B	Senior personnel: PI, 20 hrs/wk × 9 months × $10,000/person-month (mock rate)	$45,000
C	Fringe benefits, 30% of personnel (mock rate)	$13,500
D	Equipment (items >$10K)	$0
E	Materials and supplies: joint module hardware, e-skin coupon, thermal test rig, consumables	$38,000
F/G	Consultants: controls/actuation specialist + materials/e-skin specialist	$40,000
H	Other direct: I-Corps training $25,000 + TABA (commercialization assistance) $6,500	$31,500
I	Indirect costs, 25% of modified total direct costs (mock rate)	$42,000
J	Fee (profit), 7% of total direct + indirect = 0.07 × $210,000	$14,700
	TOTAL	**$224,700**

Arithmetic: direct 45,000+13,500+0+38,000+40,000+31,500 = 168,000; + indirect 42,000 = 210,000; + fee 0.07×210,000 = 14,700; total 224,700. Headroom to the $305,000 ceiling: $80,300 - the real draft either scales scope or shows the reviewer deliberate restraint. Every rate shown is MOCK pending real rates and her ruling; the fee cap (7%) and the ceiling ($305,000) are solicitation facts.

CURRENT AND PENDING (OTHER) SUPPORT

For the PI: the founder's time commitments. Mock shows the PI at 20 hrs/wk on this project; any other commitments listed or none.

FACILITIES, EQUIPMENT AND OTHER RESOURCES

The lab's compute and version-control infrastructure, the build-corpus record, and - as it forms - bench space and the machine tools the Phase II prototype needs. Named collaborator facilities land here as they onboard.

OPTIONAL / CONDITIONAL PIECES

- Letters of support (optional, strengthen the commercial section - collaborator and university letters).
- Mentoring plan: only required if a subaward funds postdocs - not currently in our shape.
- IP rights agreement: STTR-only; our SBIR shape does not require one.
- Data management plan: confirm current treatment under NSF 26-510 at drafting time.

THE KEY - HOW WE GET THIS IN

1. The timeline, stated flat

- The pitch gets submitted before November - that is the thing going in the door now, at your word.
- NSF answers pitches in about 1-2 months; the invitation is then valid for the next two deadlines.
- Full-proposal deadlines: Nov 4, 2026 / March 4, 2027 / July 7, 2027. A fresh pitch cannot make Nov 4 - the real full-proposal target is March 4, 2027, backup July 7, 2027.
- Pitch rules: one per window, max two per 12 months, max three ever for the same project; editable only the same calendar day it is submitted.

2. The steps, in order

1. **Company paper confirm (you, 5 min).** ∞ stands. Ruled today: eligibility attested as a US small business. Still open, yours: the "WA" expansion.
2. **Create the MyWork account (5 min).** nsfgov.my.site.com/mywork → "Create an account." Only registration the pitch needs.
3. **Paste and submit the pitch (30 min).** Four answers paste-ready at scratch/sbir-pitch/pitch-answers.txt (limits: 3,500 / 3,500 / 1,750 / 1,750 - ours run 2,045 / 2,341 / 1,635 / 1,562). One sitting, pick the topic from the dropdown, submit, screenshot the confirmation.
4. **Start SAM.gov now, in parallel (free, yours to fire).** Needed for the full proposal, not the pitch; entity validation takes weeks, which is why it starts now.
5. **While the pitch is in review:** onboard the two named specialists (controls, e-skin) off the ZYK-411 paper; pre-build this proposal's real draft so the invitation converts fast.

3. What makes it pass (NSF's own bar)

- Feasibility must be unproven at project start - no straightforward engineering. Ours: home-capable human-experience humanoid at one-tenth to one-third of field cost has never been demonstrated.
- Market must be specific: the home, entered through care and disability.
- Team must show the right to win: the version-controlled record, the method, and named specialist hires. PI at 20 hrs/week.

4. Money and shape (both documents)

- Phase I: up to $305,000, 6-18 months, zero equity. Phase II: up to $1,250,000, ~24 months. Fee (profit) capped at 7% of total costs. Historical Phase I funding rate 10-20%.
- Full proposal: Project Description 10 pages minimum, 15 maximum, with the required headers "Technical Objectives, Approach and Work Plan" and "Broader Impacts"; budget justification 2 pages; biosketches; current-and-pending; facilities. All shown mocked above.

5. Ruled and open

- **Ruled:** small business eligibility attested (you, Oct 23); founder surface name Cat Robinson-Rutella on business material; submission is your hand, always.
- **Open, yours:** the "WA" expansion; the topic-dropdown pick at the form (screenshot it if it looks off, we pick together).

Carrier artifacts: ZYK-486 the pitch scaffold · ZYK-476 the partnership prospectus · ZYK-410/411 grants roadmap + collaborator paper · pitch answers at scratch/sbir-pitch/pitch-answers.txt.